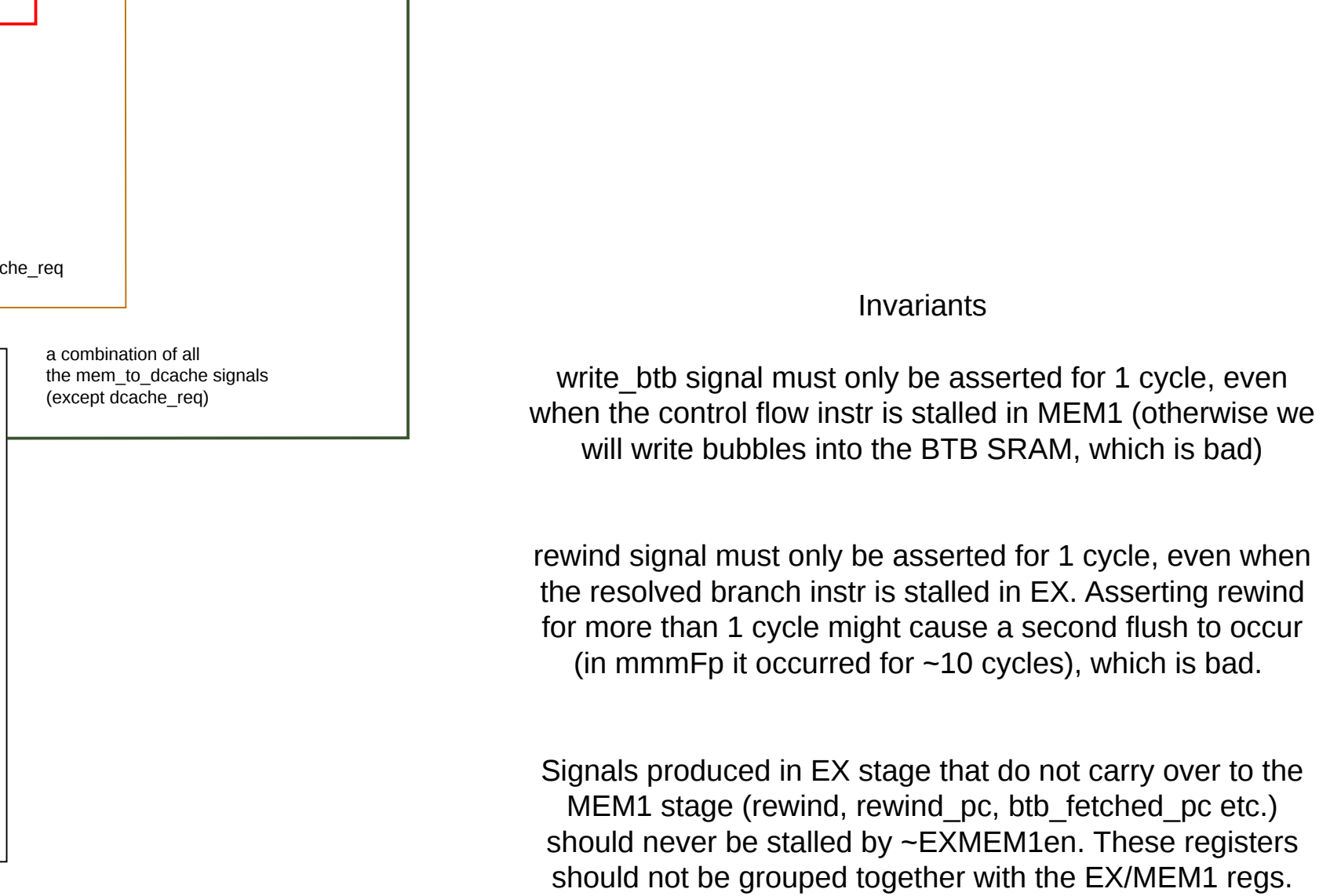
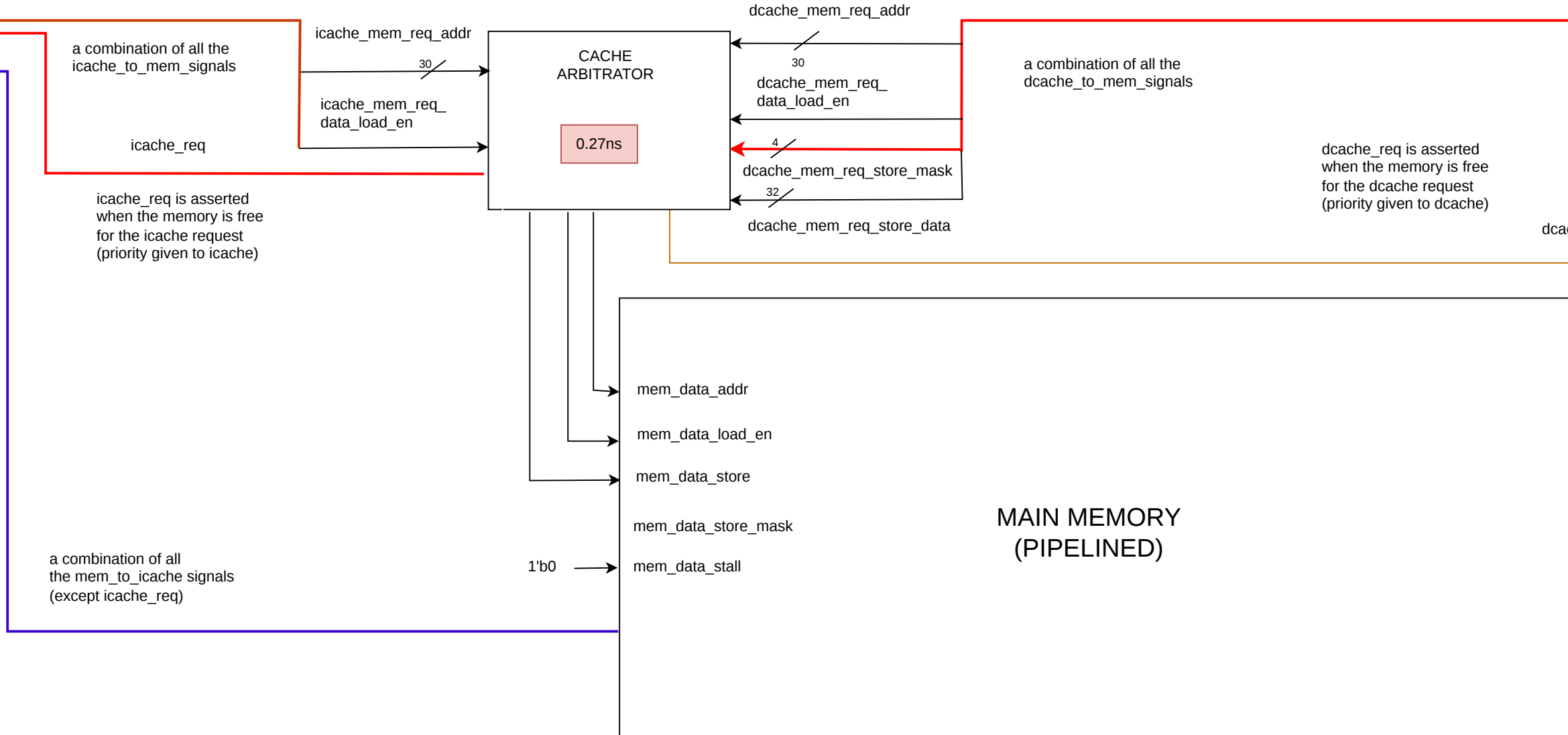
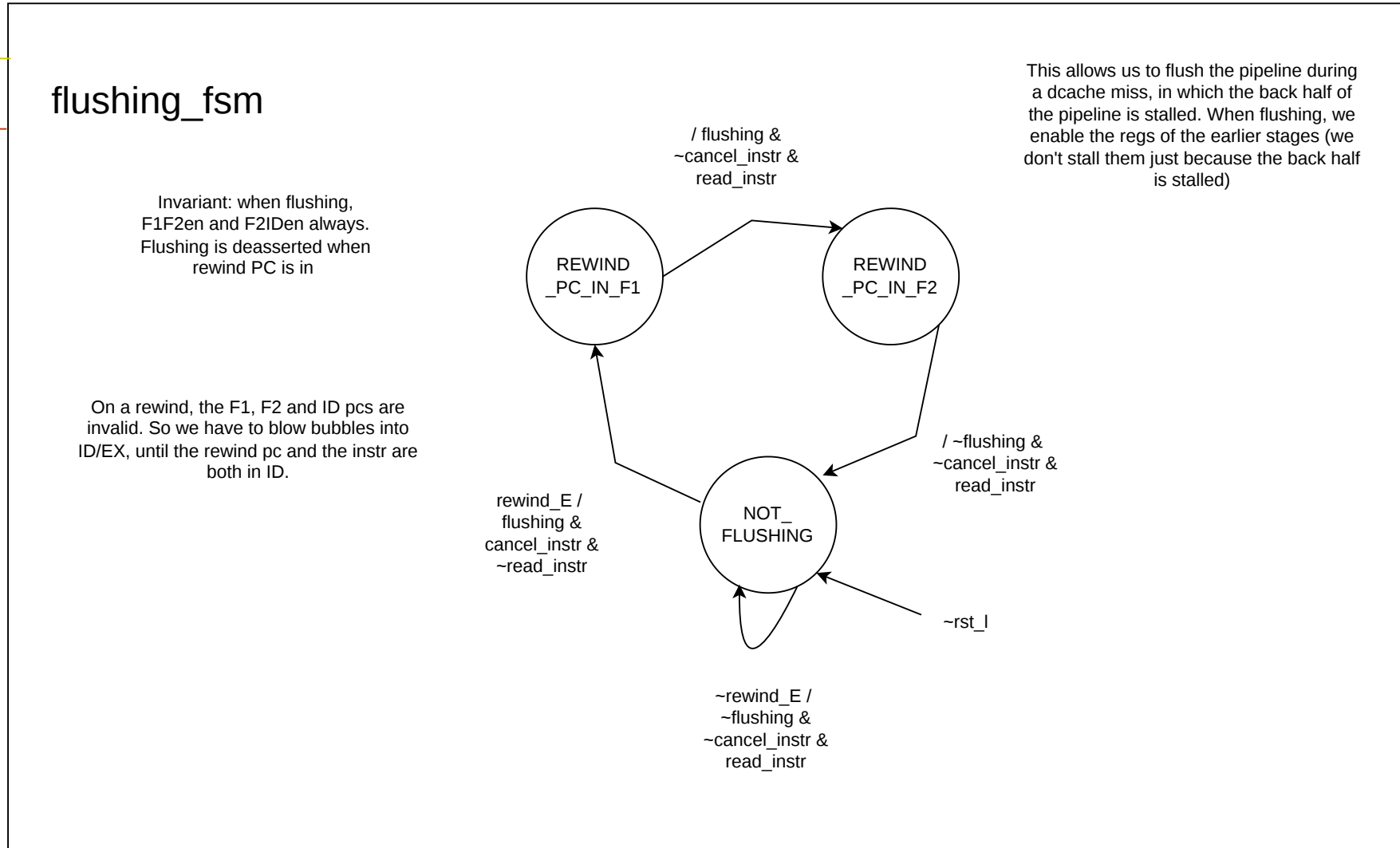
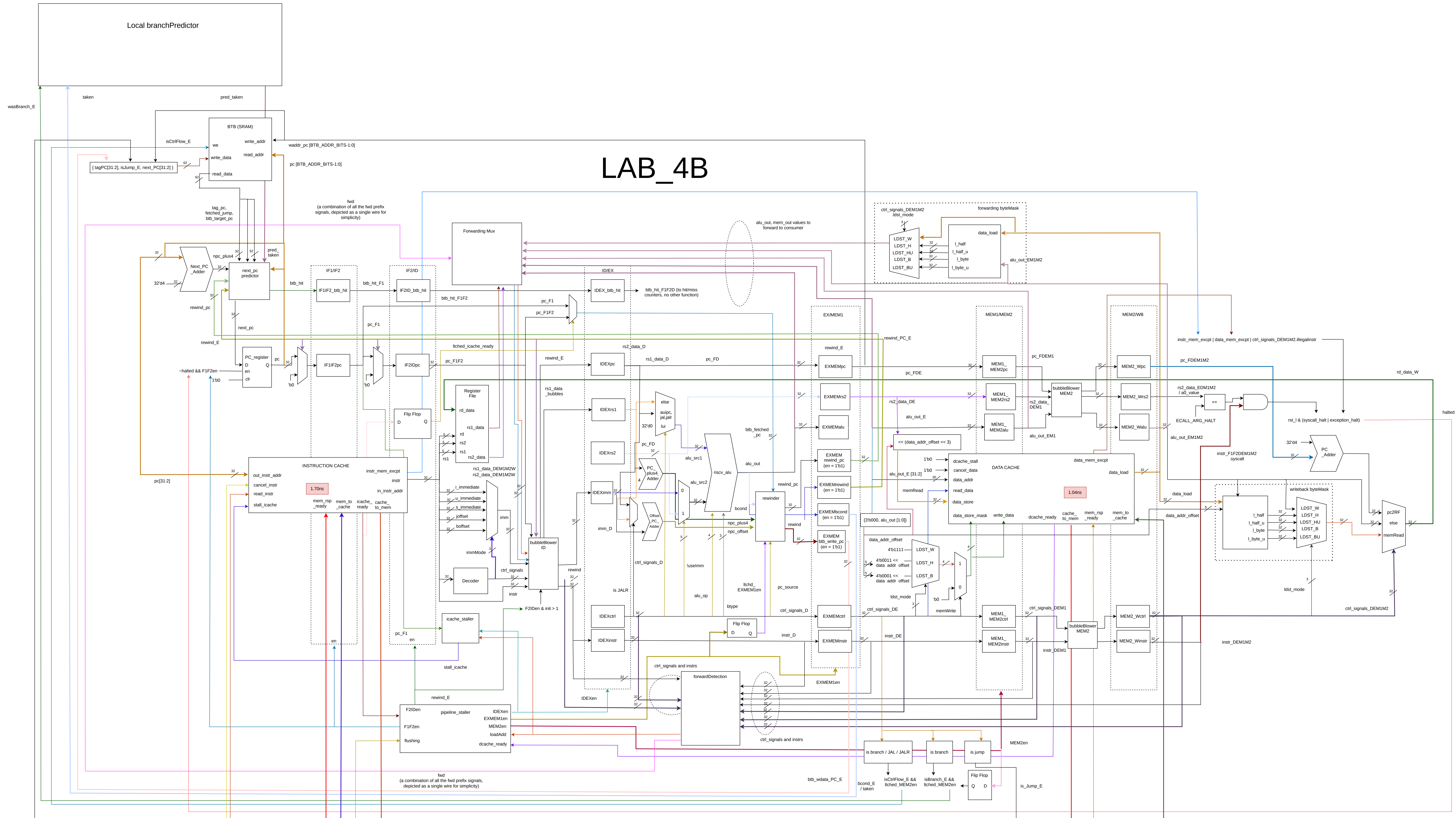


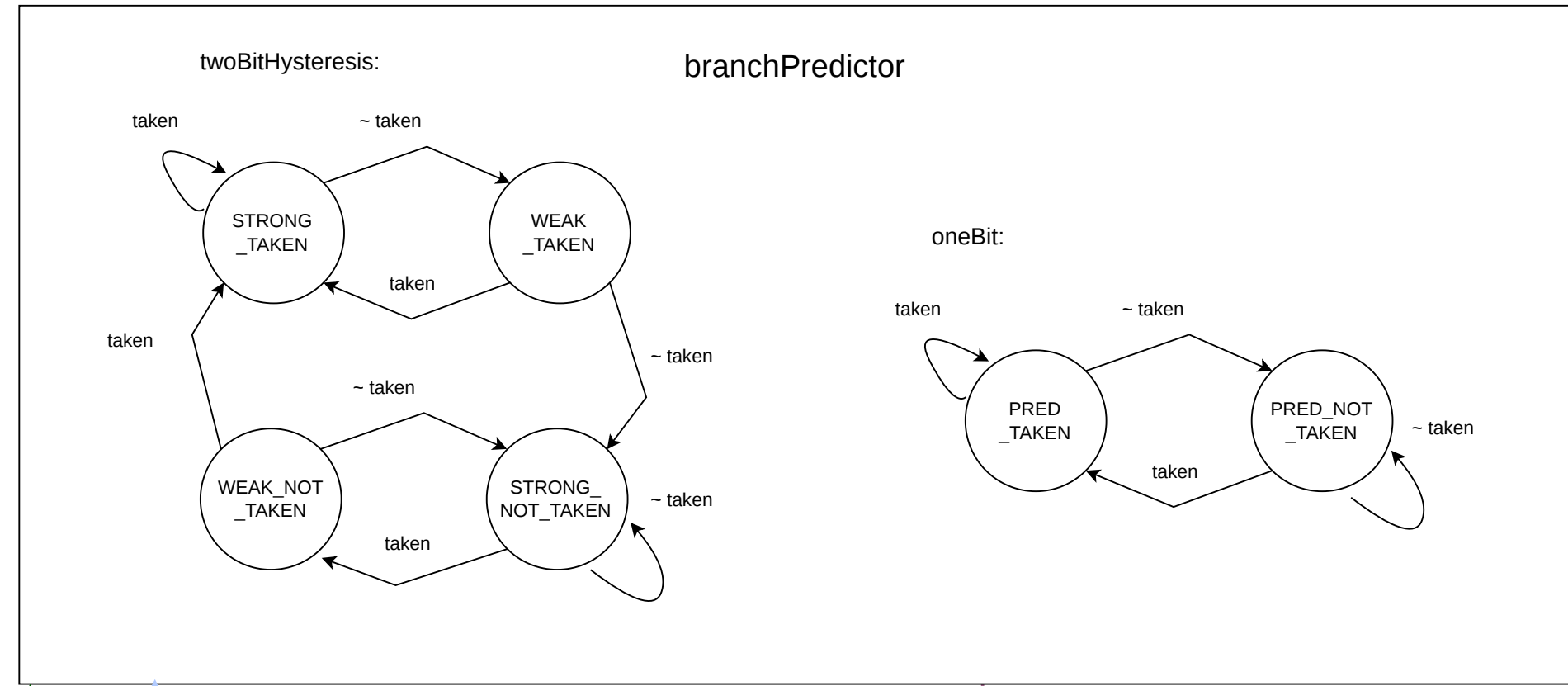


Invariants

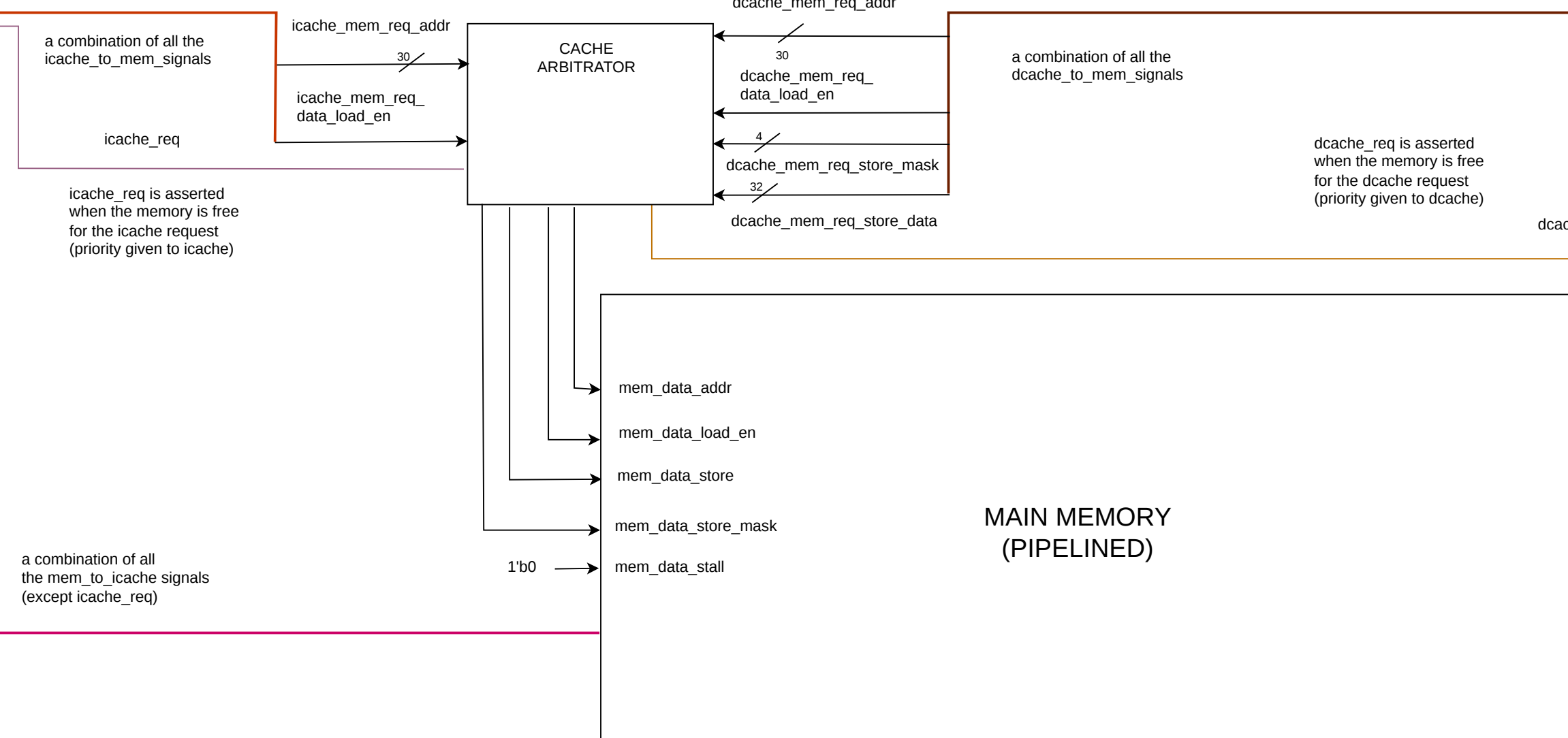
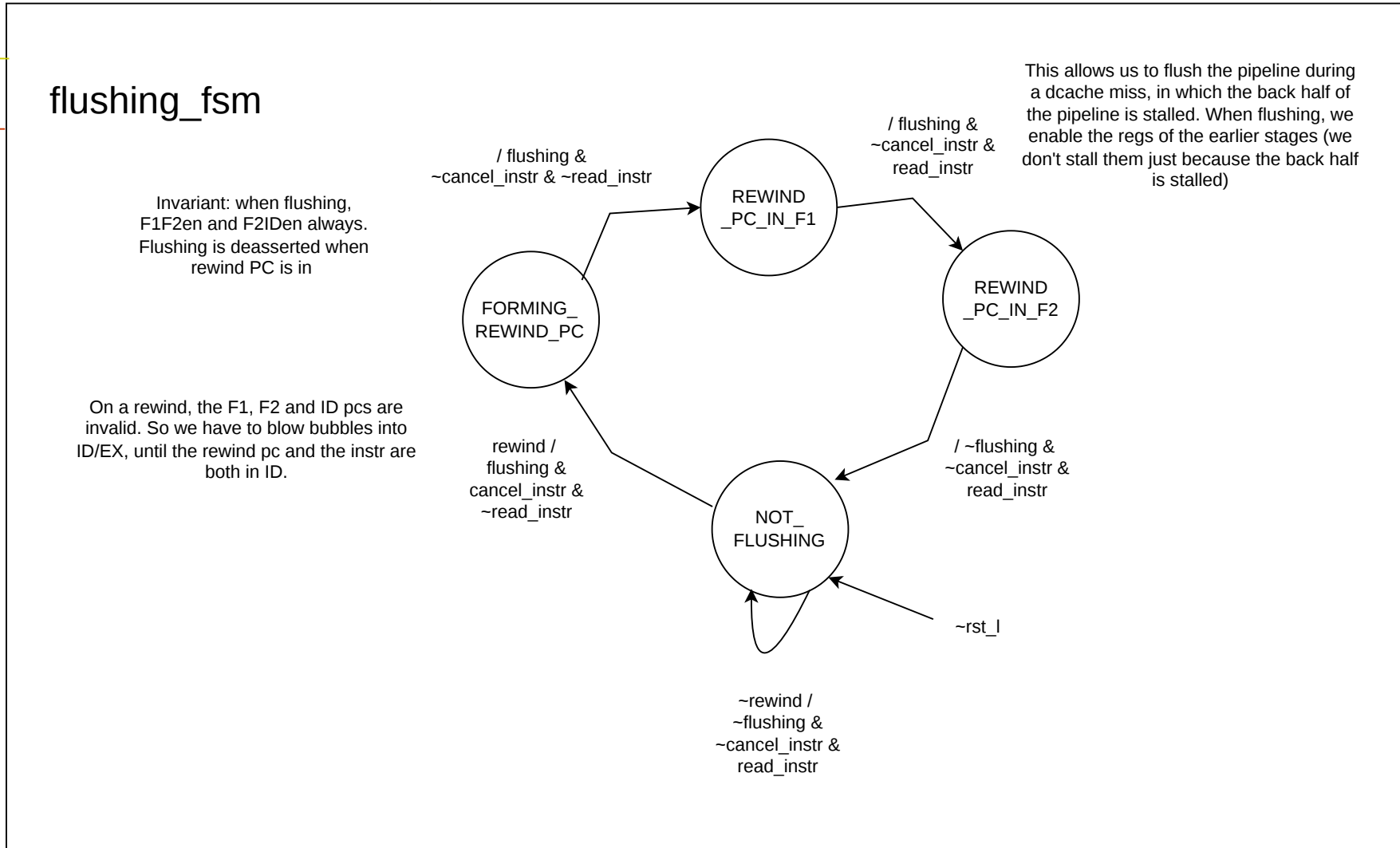
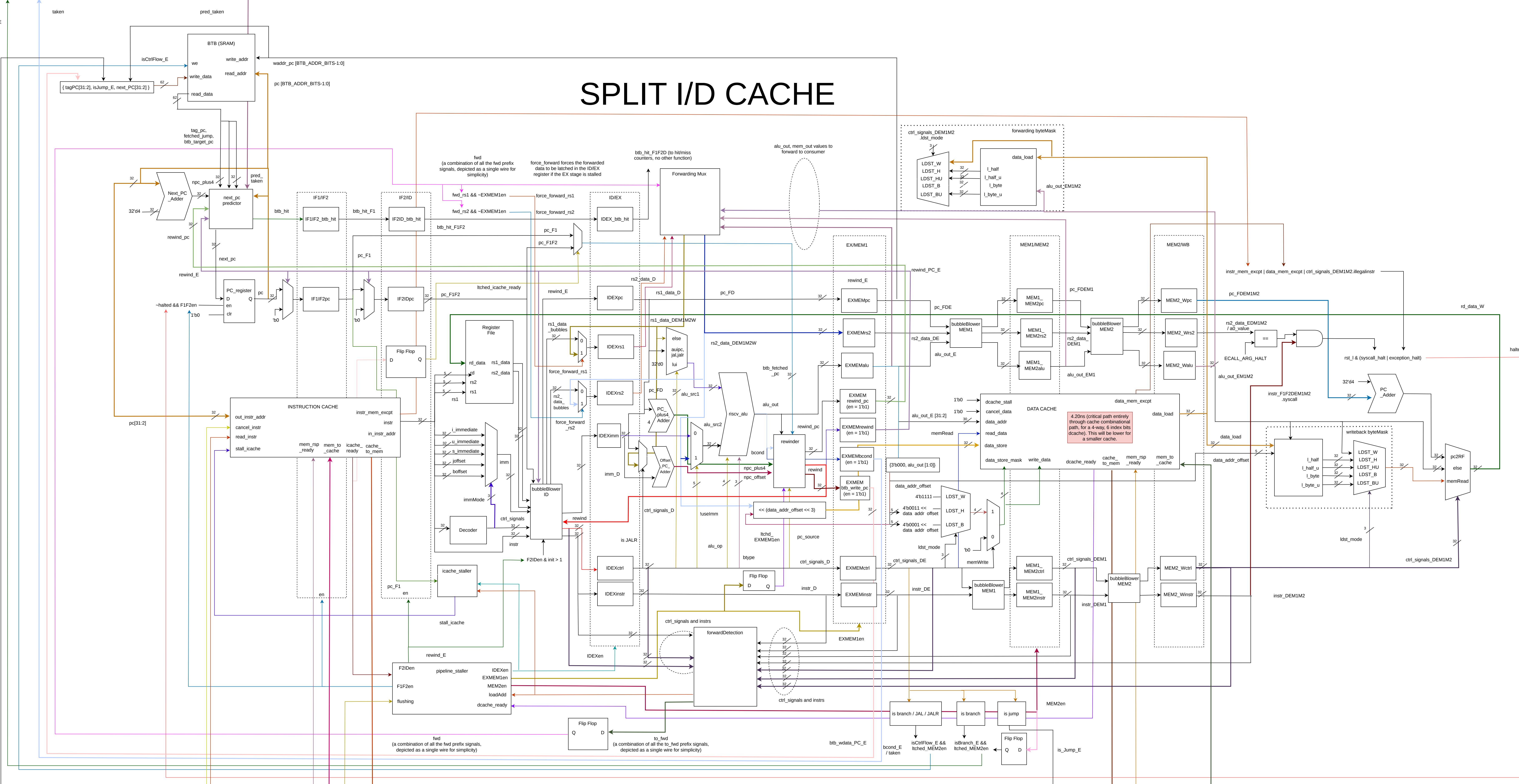
write_btb signal must only be asserted for 1 cycle, even when the control flow instr is stalled in MEM1 (otherwise we will write bubbles into the BTB SRAM, which is bad)

rewind signal must only be asserted for 1 cycle, even when the resolved branch instr is stalled in EX. Asserting rewind for more than 1 cycle might cause a second flush to occur (in mmmFp it occurred for ~10 cycles), which is bad.

Signals produced in EX stage that do not carry over to the MEM1 stage (rewind, rewind_pc, btb_fetched_pc etc.) should never be stalled by ~EXMEM1en. These registers should not be grouped together with the EX/MEM1 regs.



SPLIT I/D CACHE

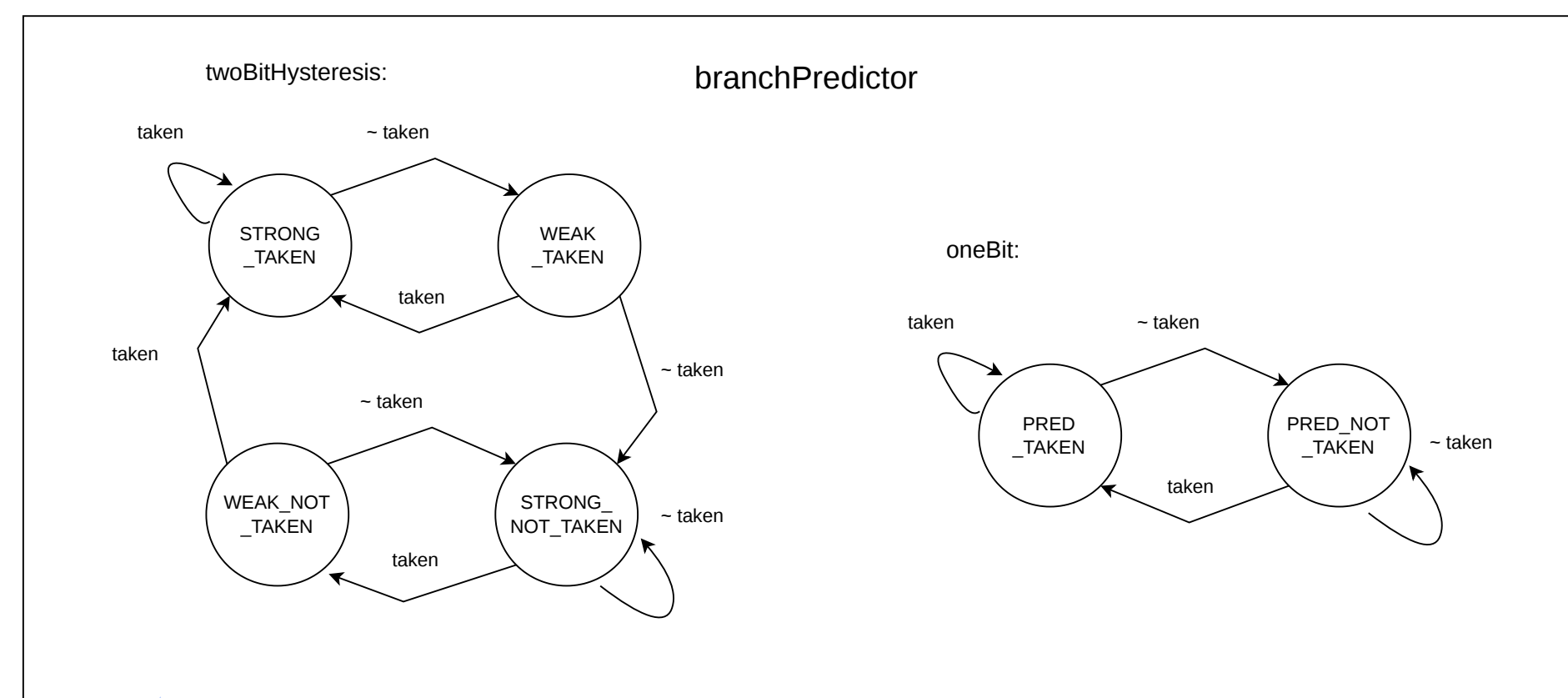


Invariants

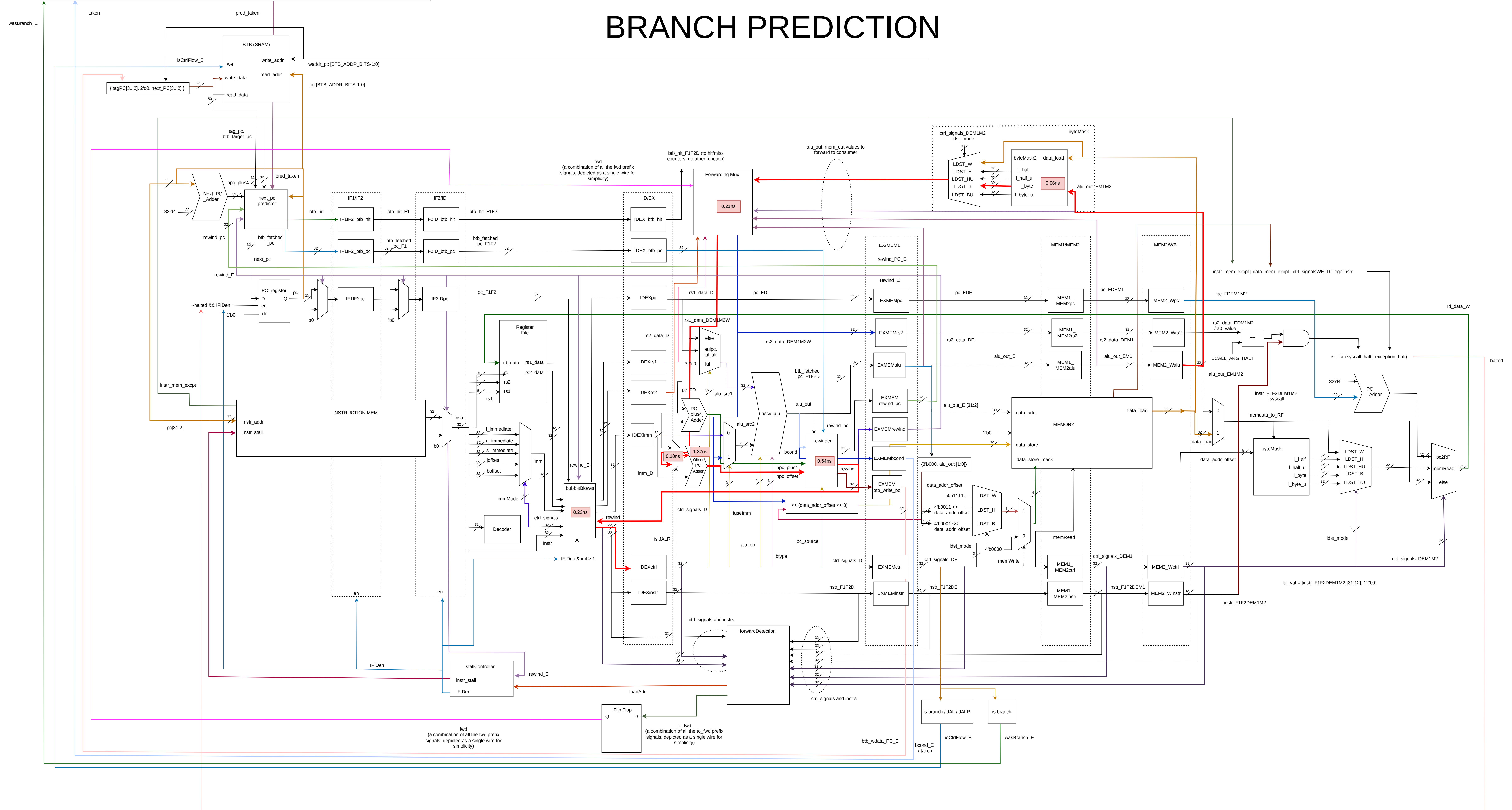
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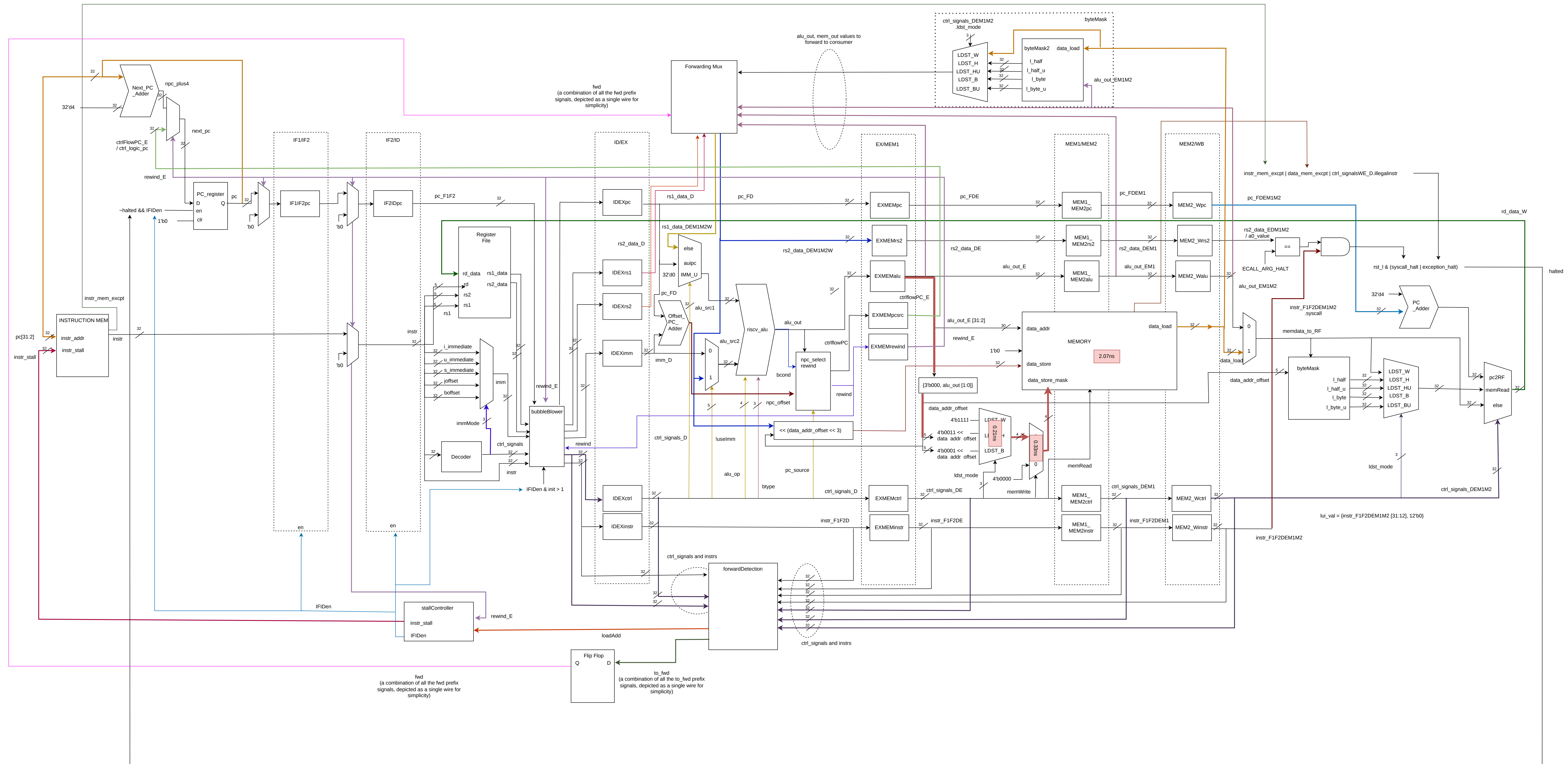
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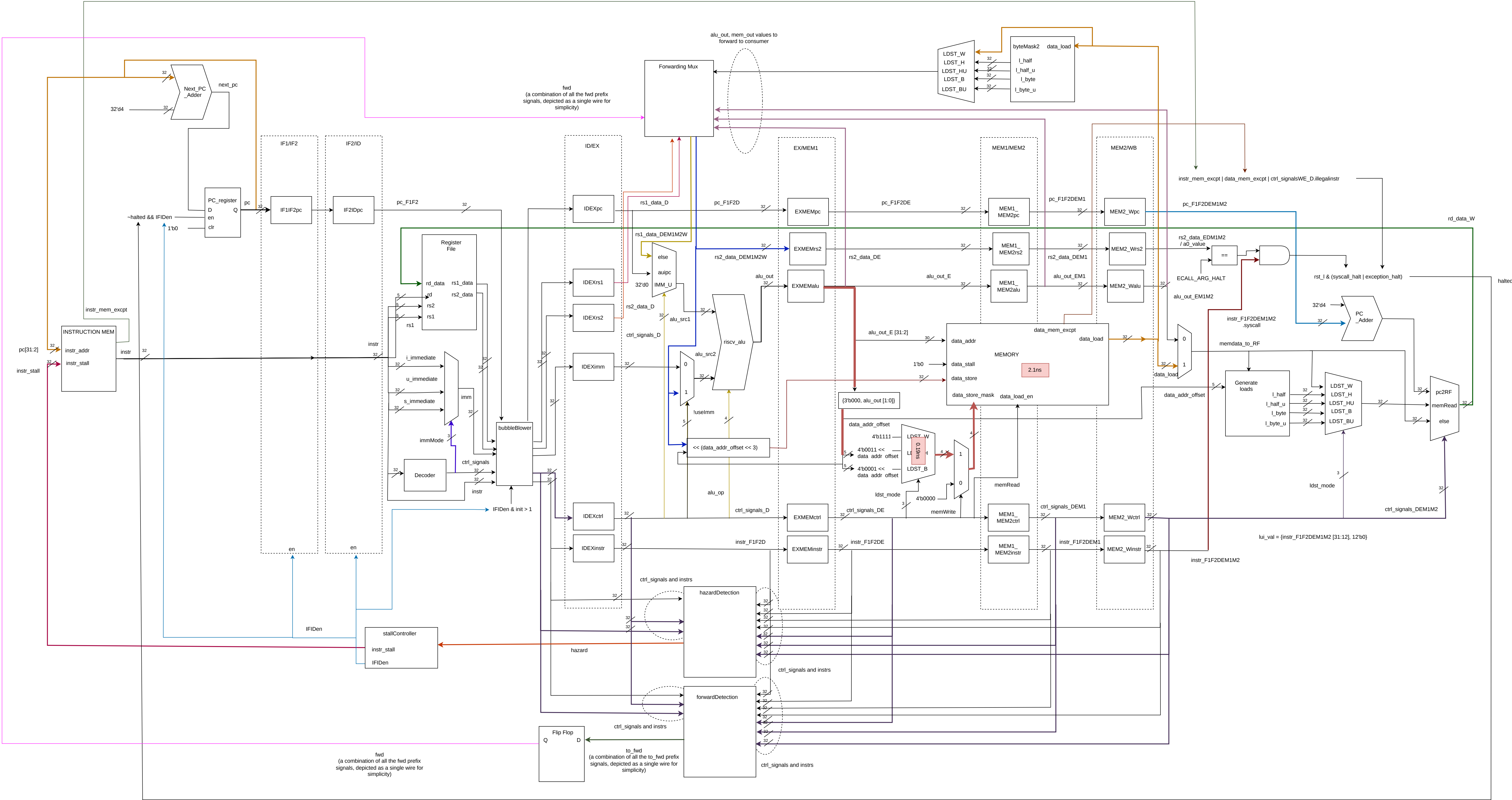
BRANCH PREDICTION



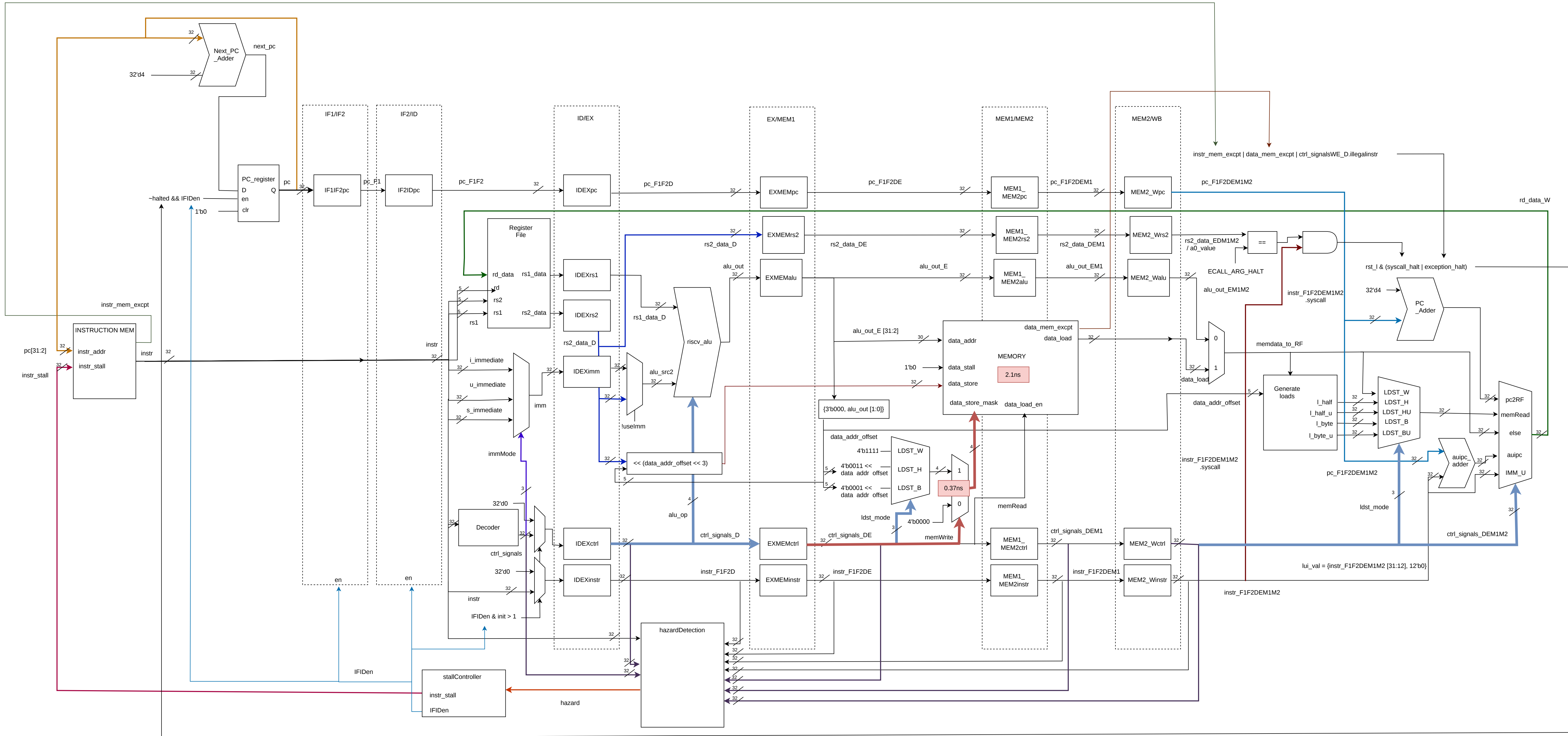
CONTROL FLOW



FORWARDING



STALL FINAL



SINGLE CYCLE

